Advances in Electrical Grid Assets Inspection: Exploring Multimodal Large Language Models

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***Abstract*—Electrical grid asset inspection is crucial for ensuring infrastructure reliability, preventing failures, and optimizing maintenance strategies. Traditional methods rely heavily on manual labor, making inspections time-consuming, costly, and prone to inconsistencies. Recent advances in deep learning, particularly CNNs and ViTs, have shown potential for automating visual inspection tasks. Vision models are evolving towards multimodal, large-scale architectures. While MLLMs trained on diverse internet-sourced datasets show strong performance, many industrial tasks require models fine-tuned on proprietary data to capture the nuanced features necessary for specialization.**

This work explores the capabilities and limitations of multimodal large language models for electrical grid asset inspection, comparing pre-trained and fine-tuned versions of the Florence-VL model across tasks like detailed captioning, object detection, and segmentation. Ground truth labels for caption-related tasks were generated using GPT-4, while annotations from a dataset of insulator defects were used for region-related tasks. Results indicate that fine-tuning improves class recognition and object localization, though captions often remain simplistic and lack domain-specific detail. Despite GPT-4 providing accurate captions, it lacks batch-processing capabilities, limiting its scalability. These findings underscore the importance of domain-specific fine-tuning and high-quality data to enhance MLLM performance.

I. Introduction

Electrical grid increasing reliability is driven by the rising dependency on electrical power. The electrical power growing demand, combined with the integration of renewable energy sources and distributed generation, not only increases the electrical grid management complexity but also requires a significant expansion of the grid in the near future, intensifying the demand for efficient maintenance strategies.

Automating visual inspections improves efficiency and consistency by addressing the limitations of manual inspections. Manual checks require skilled workers, and are labor-intensive and time-consuming. Human inspectors inconsistent judgments, promoted by fatigue and cognitive bias, can lead to errors and variability [1]. While automated systems help reduce costs and improve consistency, they come with their own expenses and challenges [2], such as, development and maintenance costs, integration with existing systems efforts, or service failure handling overload. Additionally, achieving accuracy beyond human capabilities remains a challenge.

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Recent advances in computer vision have been driven by Deep Learning (DL) models, which have shown impressive robustness to significant label noise, especially when trained on large datasets. However, even with abundant data, models trained on clean labels consistently outperform those trained on noisy labels [3]. Despite the critical impact of data quality, many studies adopt a model-centric approach, focusing on refining neural network architectures or learning algorithms [4]. While this has led to breakthroughs in general-purpose computer vision, it often overlooks the foundational role of high-quality data. In domains with limited or specialized datasets, a data-centric approach, prioritizing carefully curated data with accurate annotations, can effectively guide the learning process [5]. Neglecting this aspect may lead to solutions that fail to meet the specific demands of industrial applications.

Multimodal Large Language Models (MLLMs) have demonstrated remarkable performance in generic tasks, such as image-to-image translation [6], but their application in industry-specific contexts, such as electrical grid inspections or photovoltaic defect detection [7], remains limited. Generic training datasets, like ImageNet or COCO, fail to capture the nuanced characteristics of industrial images, which often feature repetitive patterns, subtle defects, or low-contrast regions.

To address these challenges, this work makes several contributions towards adapting MLLMs for electrical grid asset inspection. The paper explores the potential of Florence-VL for improving spatial granularity through region-to-segmentation tasks, while also identifying limitations in domain-specific applications, such as performance constraints in semantic understanding. It also shares a curated version of the EPRI Insulator Defect Image Dataset [8], standardizing file structures, removing redundant augmentations, and manually annotating the test split to enable consistent performance evaluation.

II. Evolution of Deep Vision

Deep learning for vision has evolved from Convolutional Neural Networks (CNNs) to Vision Transformers (ViTs) and Large Vision Models (LVMs). CNNs like AlexNet and ResNet transformed feature extraction, forming the backbone of defect detection systems. ViTs introduced self-attention for long-range dependencies, while hybrid models improved scalability. More recently, models like SAM and Florence leverage large-scale training and multimodal learning, enabling zero-shot generalization and text-based prompting. This evolution unlocks new possibilities for grid asset inspection, where general-purpose models adapt to complex, real-world scenarios.

A. Convolutional Neural Networks

CNNs have been foundational in Computer Vision (CV) tasks such as Object Detection (OD). Their ability to hierarchically extract features has been extensively validated in tasks handling structured data distributions, such as images. A typical CNN model has two main components: a feature extractor, composed of convolutional layers, and a classifier, usually consisting of dense layers. As the network advances, spatial dimensions shrink while depth increases, transforming a high-dimensional matrix into a reduced vector.

The history of CNNs began in 1998 with the introduction of LeNet-5 [9], a pioneering architecture composed of 5 layers, 3 convolutional and 2 fully connected. In 2012, AlexNet [10], marked a significant leap in computer vision by classifying 1.2 million mid-resolution images into 1,000 categories with 37.5% top-1 error rate, and winning the ImageNet competition.

VGG16 [11], developed in 2014, featured a deeper CNN architecture, housing 138 M parameters across 13 conv layers and 3 dense layers. That same year, the Inception [12] architecture from Google introduced the inception module, which efficiently captures multiscale features through parallel convolutions and pooling. This enabled a 22-layer NN with only 13 M weights, improving accuracy with fewer parameters.

ResNet [13], introduced by Microsoft in 2015, revolutionized DL by introducing residual connections, addressing the vanishing gradient problem in deep networks, and enabling effective training of NN with hundreds of layers. In 2017, DenseNet [14] set a new SOTA by replacing sequential residual connections with dense ones, reducing the need for deep networks and achieving better results with fewer parameters.

In 2020, EfficientNet [15] established new benchmarks with the introduction of an efficient compound scaling coefficient. Acknowledging that NN scaling occurs in three dimensions (image size, width, and depth), they developed a method for computing the optimal trade-offs between these dimensions, resulting in better architectural choices.

ConvNeXt [16] further improved the state-of-the-art by selecting and combining several previously introduced concepts, including augmentation, regularization, kernel size, optimizer, activation, and layer normalization. In practice, it involved a grid search of hyperparameters, incorporating those reported in recent vision transformer models.

B. Vision Transformers

Transformers marked a significant shift in DL methodologies. Originally developed for Natural Language Processing (NLP), they outperformed traditional RNNs [17], which struggled to capture long-range dependencies. The paradigm shift occurred with the introduction of the Transformer architecture [18], which replaced recurrence with self-attention, enabling efficient parallel processing and improved context modeling. Self-attention computes similarity scores between input embeddings to capture long-range dependencies. This mechanism was later extended [19] to incorporate relative positional information, further enhancing contextual understanding.

This innovation enabled language models like BERT [20], which used bidirectional encoder representations to generate contextualized embeddings, achieving state-of-the-art results in NLP tasks like question answering and sentiment analysis.

Vision Transformers emerged in 2020 [21], introducing a novel approach to image analysis by dividing images into fixed-size patches (e.g., 16x16), embedding them as tokens, and applying multi-head self-attention mechanisms, similar to how words are processed in NLP. Unlike CNNs, which rely on local feature hierarchies, ViTs capture global dependencies through self-attention. Google's ViT [21] outperformed CNNs on the ImageNet benchmark but required pretraining on 300 million images instead of the standard 1.2 million. Later that year, Meta introduced the Data-efficient Image Transformer (DeiT) [22] to address the data-hungry nature of ViTs. DeiT incorporated a teacher-student training paradigm, where a CNN encoder generated a knowledge distillation token to guide the Transformer student in learning local features representations. This improved ViT efficiency, enhanced generalization, and reduced reliance on large datasets, making them more practical for real-world applications.

In 2021, the Swin Transformer [23] improved the ViT architecture by reintroducing ConvNet priors through the Shifted Windows scheme. This approach divided the image into non-overlapping windows and applied self-attention within each window, while shifting the windows across layers to capture long-range dependencies. The modification enhanced computational efficiency and scalability for high-resolution images, setting new benchmarks in computer vision tasks like image classification, object detection, and segmentation.

C. Foundation Models & Multimodal LLMs

Foundation models [24] have advanced AI by integrating modalities like vision, text, and audio, enabling sophisticated contextual reasoning and task generalization. Vision-language (VL) models, such as CLIP and SAM, have played a key role in pushing the boundaries of computer vision. CLIP (Contrastive Language-Image Pretraining) [25], introduced by OpenAI in 2021, uses contrastive learning to map images to textual descriptions, enabling zero-shot capabilities in tasks like classification and detection. This approach allows it to be applied across various tasks without task-specific training.

SAM (Segment Anything Model) [26], developed by Meta in 2023, focuses on class-agnostic image segmentation. Trained on over 1 billion masks from 11 million images, it sets new benchmarks for flexibility and zero-shot performance across tasks. Its promptable design adapts easily to new scenarios, often outperforming prior models.

In August 2023, Alibaba introduced Qwen-VL [27], a vision-language model for tasks like text reading, localization, image captioning, and question answering, setting new benchmarks in vision-centric tasks with a multilingual, multimodal cleaned corpus. In September 2023, OpenAI released GPT-4V [28], a multimodal model that combines computer vision and language understanding, enabling the processing of both text and visual inputs for enhanced data interpretation.

Meta released Llama 3 [29] in November 2024, a multimodal foundation model with 405 billion parameters, excelling in tasks like image, video, and speech recognition. Llama 3.2 Vision [30] integrates visual capabilities for multimodal tasks, including image analysis. In December 2024, Google DeepMind published PaliGemma 2 [31], an upgraded Vision-Language Model (VLM) from the Gemma 2 family, excelling in tasks such as table structuring, molecular structure recognition, and radiography report generation.

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Florence-VL [32], released by Microsoft Research in December 2024, builds upon the Florence-2 model by integrating enriched visual features through a novel Depth-Breadth Fusion (DBFusion) architecture. This approach combines visual features from different depths and breadths, aligning them with pretrained large language models like Phi 3.5 and Llama 3. Florence-VL excels in various multimodal tasks, including Visual Question Answering (VQA), OCR, chart analysis, and knowledge-intensive tasks. By leveraging end-to-end pretraining on high-quality image captions and instruction-tuning pairs, it outperforms previous models on multiple benchmarks. Florence-VL's open-source availability encourages further research and innovation, making it a valuable tool for real-world applications that require both visual and textual understanding.

III. Methodology

We evaluated the Pre-Trained (PT) and Fine-Tuned (FT) versions of the Florence-VL model [32] on images of insulator defects, reporting both quantitative metrics and qualitative examples. The experiments cover captioning, object detection, and region-based segmentation, with model prompts accepting images, text descriptions, or region coordinates as inputs.

The fine-tuning process was carried out using Low-Rank Adaptation (LoRA) [33], a lightweight technique that significantly reduces the number of trainable parameters, enabling training on standard hardware. Experiments were conducted on a workstation with a 16-core CPU, 256 GB of RAM, and an NVIDIA RTX A6000 GPU (48 GB VRAM). Several hyperparameter combinations were tested with similar results. The FT version presented here was trained on original image-size data for 100 epochs, using a learning rate of 1×10^{-5}, batch size of 8, LoRA alpha of 8, dropout of 0.05, and LoRA rank of 16, resulting in approximately 3.9 million trainable parameters, representing 1.4% of the total model weights.

A. Experimental Design

The first set of experiments receives the image as input, along with a prompt specifying one of the following tasks: `'<CAPTION>'`, `'<DETAILED CAPTION>'`, or `'<MORE DETAILED CAPTION>'`, and outputs a text sentence.

The second set of experiments also uses the image as input but returns structured data, such as coordinates represented by numerical values. Task prompts include `'<REGION PROPOSAL>'`, which outputs bounding boxes of object likeness; `'<OD>'`, which also includes the object's class along each bounding box in its response; and `'<DENSE REGION CAPTION>'`, which generates a sentence caption for each detected object instead of its class.

The third set of experiments prompts not only the image and task but also a textual input. The `'<CAPTION TO PHRASE GROUNDING>'` task returns objects with captions matching the provided phrase. The `'<REFERRING EXPRESSION SEGMENTATION>'` task outputs polygon points corresponding to the input expression. Finally, the `'<REGION TO SEGMENTATION>'` task converts bounding boxes to fitted polygons.

The ground truth labels for caption tasks were provided by the GPT-4o, given its state-of-the-art language capabilities. For region-related tasks, the labels were sourced from IDID [8], which were also used to fine-tune the Florence-VL model [32].

B. Dataset Description

To conduct our experiments, we used the publicly available Insulator Defect Image Dataset (IDID) [8], which contains high-resolution, close-up images from real operational environments. Since the IDID [8] was shared in a competition context, the test set annotations were kept private. As a result, the test set contains 88 unlabeled images, while the training dataset includes 1,600 annotated images across four classes: String Insulator, Good Disc, Broken Disc, and Flashover Shell. However, only 400 images are unique, with the remaining images generated through offline mirroring augmentation, which may require additional care in how the validation is performed.

These dataset characteristics, among others, were carefully identified in the study conducted in [34], leading to the careful selection of images for the validation set, while maintaining the original 1,600 image files. However, removing redundant data is helpful for expediting the training process. Building upon previous work [34], we not only eliminated duplicate or unnecessary augmented data to streamline dataset splitting but also standardized file names and converted labels into a consistent format. We further enriched the dataset by adding annotations to the test set, enabling quantitative performance evaluation, which was not feasible with the data used in [34]. Importantly, no modifications were made to the image content or label accuracy of the originally annotated training and development sets.

We also generated multiple versions of the dataset, applying resizing with zero-padding to preserve the aspect ratio and resizing the images to common dimensions (1280^2, 640^2), ensuring compatibility with existing models and standard hardware configurations. Finally, we made this version of the dataset publicly available through the Roboflow [35] platform, giving proper credit to the original authors and facilitating reproducibility, enabling comparable results and benchmarks.

Figure 1 presents canonical samples of each class, while Figure 2 illustrates the class distribution across splits sets.

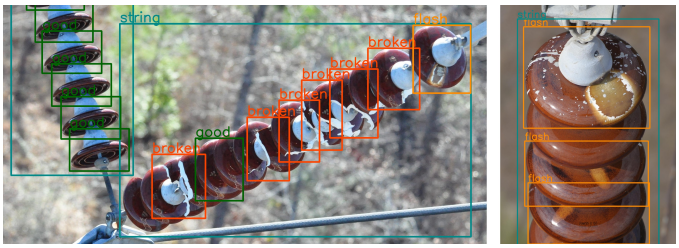


Fig. 1. Canonical samples of each class. "String" represents the chain of cap and pin insulators. "Good" represents insulator discs without defects. "Broken" refers to insulator discs with sharpened shape, core exposure, or disc absence. "Flash" refers to discs with damaged coating and lighter marks or white bands.

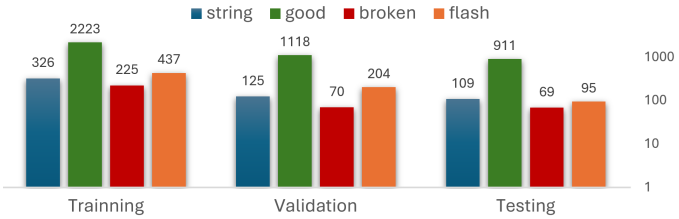


Fig. 2. Class distribution across dataset splits. The training set (300 images) contains 3,211 instances, the validation set (100 images) has 1,517, and the testing set (88 images) includes 1,184. On average, "string" accounts for 9.5%, "good" 71.9%, "broken" 6.2%, and "flash" 12.4% of the total instances.

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IV. Results and Discussion

The results of the *Detailed Caption* task for the Pre-Trained (PT) and Fine-Tuned (FT) models, presented in Tables I and II, respectively, for the images in Fig. 3 and Fig. 4, illustrate the performance of both models, with a comparison to GPT-4.

The PT Florence-VL model generated relevant terms like, "rust", and "wire", but also incorrect labels, such as "beads" and "candleholder", due to its limited vocabulary. After fine-tuning on the EPRI insulator defect dataset [35], the FT model learned relevant classes but still produced simplistic, often unreadable captions, suggesting the need for further domain-specific linguistic fine-tuning with caption-image pairs.

TABLE I. Results for Detailed Caption Task

Model	Inference Results for Detailed Caption of the Image in Fig. 3
PT Florence	"A tall stack of chocolate and white candies on the ground.", "There is a tall stack of brown and white balls on the ground. The ground is brown and has green grass on it. There is a small white ball in the middle of the stack of balls."
FT Florence	"goods in a line in front of a blurry background", "goods are goods. goods are browns are goods.goods have goods on them. goodgoods isgoods andgoods"
GPT4-o	"A vertical string of dark brown ceramic insulator discs with a glossy finish, separated by light gray cylindrical spacers. The insulator chain is part of a high-voltage power line, with a white, irregularly shaped flash mark or residue on one of the discs, possibly indicating a discharge or damage. The background consists of a blurred ground with patches of green vegetation and brown soil."

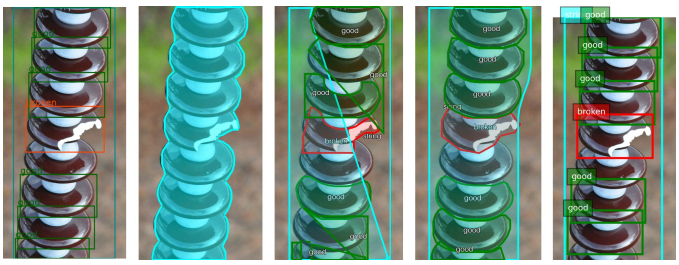


Fig. 3. Results on Pre-Trained (PT) and Fine-Tuned (FT) models. (a) Original annotations (b) Referring expression to segmentation (PT) (c) Regions to segmentation (PT) (d) Regions to segmentation (FT) (e) Object detection (FT)

The *Region Proposal* task generally performed well with a blurred background, correctly identifying the first-plane object in the PT model, while the FT model successfully proposed regions containing each insulator disc. However, caption-to-phrase grounding, using text input like "locate every defective disc, flash, or broken," showed limited results in both models.

The *Referring Expression to Segmentation* task, with input like "string containing all insulator discs, good or defective," performed well on blurred background images, even with the PT model, as shown in Fig. 3.b). However, some incorrect points were predicted, as illustrated in Fig. 4.b).

The *Region to Segmentation* task generally yielded good results. In some cases, there was a clear improvement from the PT (Fig. 3.c)) to the FT (Fig. 3.d)), while in other cases (Fig. 4.c) and Fig. 4.d)), some polygons fit the objects better after fine-tuning, while others became less accurate.

Despite poor performance in the *Object Detection* task for the PT model, the FT showed promising results, achieving a mAP50 of 53.6%. While not competitive with related benchmarks (89.7% for YOLOv8 [34] and 94.8% for YOLO11 [36]), and mismatched results in more complex cases (Fig. 4.e), it provided accurate results in simple ones (Fig. 3.e).

TABLE II. Results for Detailed Caption Task

Model	Inference Results for Detailed Caption of the Image in Fig. 4
PT Florence	"There is a metal pole with a white cord attached to it. There are brown beads on the top of the pole. There is a brown piece of metal on top of them."
FT Florence	"A string is hanging from a cord. The cord is white. There are six strings. The string is made of metal. The metal is gray. The lights are yellow. The flash is yellow."
GPT4-o	"High-voltage ceramic insulator string with multiple brownish-orange disc insulators, connected to a transmission line. The insulators show visible signs of damage, including burn marks and erosion, especially on the top disc. Yellowish deposits are present on several discs, which may indicate contamination or material degradation. The assembly is secured to a metal crossarm, which has rust and corrosion, and the conductor is attached through a steel cable.

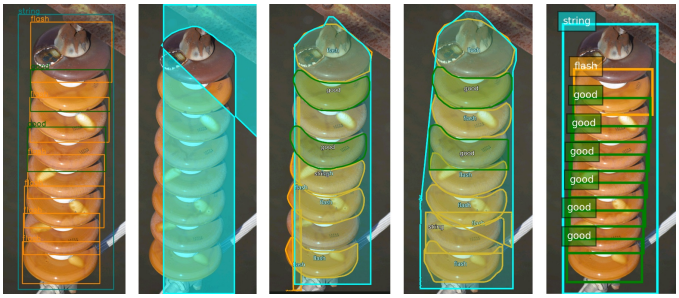


Fig. 4. Results on Pre-Trained (PT) and Fine-Tuned (FT) models. (a) Original annotations (b) Referring expression to segmentation (PT) (c) Regions to segmentation (PT) (d) Regions to segmentation (FT) (e) Object detection (FT)

V. Conclusion and Future Work

In this paper, we explored the potential and limitations of MLLMs applied to industry-specific tasks, focusing on electrical grid asset inspections, particularly defect detection in HV insulators. While vision models are evolving towards multimodal, large-scale architectures, their performance in general-purpose tasks is strong due to training on diverse Internet-sourced datasets. However, their application to specialized contexts like industrial defect detection is limited by a lack of domain-specific knowledge and tailored training data.

Through our experiments with Florence-VL models, we demonstrated solid performance in detection and segmentation tasks; however, they struggle with image captioning tasks. In contrast, GPT-4 generated more accurate and domain-relevant captions, yielding impressive results. However, its learning strategy is constrained by user interactions through the chat interface, making it less practical for handling proprietary data or batch processing tasks. Meanwhile, the limited vocabulary of the Florence-VL model emphasizes the need for further fine-tuning with industry-specific data to enhance its performance.

Besides benchmarking other available datasets containing different electrical grid assets with more SOTA models, future work should focus on developing tailored datasets with detailed captions using domain-specific vocabulary that better represent the complexities of industrial environments. This will help reduce the semantic gap between visual and textual terms and improve feature extraction in MLLMs. Additionally, the effectiveness of fine-grained spatial annotations should be validated through weakly supervised box-to-polygon conversion, facilitating efficient training of smaller CNN-based models.

Ultimately, while LVMs have advanced computer vision, their true potential in specialized industrial tasks can only be realized by addressing the limitations of generic datasets and focusing on domain-specific adaptations.

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